

MIST Master's Examination — Practice Set 2 of 4

Data Center Environment: Mixed Topics

Intelligent Storage Systems · Fibre Channel SAN · IP SAN & FCoE · Network-Attached Storage

Format: 20 MCQs (Medium–Hard) · Structural Questions (30 Marks) · Essay Questions (50 Marks)

Coverage in this set: Cache protection & high-end/midrange arrays (Ch.4) · FC topologies & classes of service (Ch.5) · FCoE architecture (Ch.6) · Unified/gateway NAS connectivity & performance tuning (Ch.7)

SECTION A: Multiple Choice Questions (20 Questions)

1. (Medium) [Ch.4] Which cache management algorithm frees the pages that have been accessed MOST recently, on the assumption that recently used data is unlikely to be needed again soon?

- A. Least Recently Used (LRU)
 - B. Most Recently Used (MRU)
 - C. First In First Out (FIFO)
 - D. Clock algorithm
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2. (Hard) [Ch.4] Which type of flushing occurs in the event of a large I/O burst when cache reaches 100% of its capacity, significantly affecting I/O response time?

- A. Idle flushing
 - B. High watermark flushing
 - C. Forced flushing
 - D. Background flushing
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3. (Medium) [Ch.5] In a core-edge fabric topology, which tier is typically composed of enterprise directors and ensures high fabric availability, with all storage devices connected to it?

- A. Edge tier
 - B. Core tier
 - C. Access tier
 - D. Distribution tier
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4. (Hard) [Ch.5] What does "hop count" represent in a core-edge fabric, and what is the recommended best practice regarding it?

- A. The number of physical disks in a RAID set; should be minimized to 1
- B. The number of ISLs traversed by a packet between source and destination; best practice is to keep host-to-storage hops at 1

- C. The number of zones configured in the fabric; should equal the number of hosts
 - D. The number of VSANs created; should not exceed 4
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5. (Medium) [Ch.6] What is the primary benefit of FCoE (Fibre Channel over Ethernet) in a data center?

- A. It eliminates the need for any storage networking protocol
 - B. It consolidates LAN and SAN traffic over a single physical interface infrastructure, reducing adapters, cables, and switches
 - C. It replaces TCP/IP entirely with FC
 - D. It increases the number of FC zones supported
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6. (Hard) [Ch.6] What does a Converged Network Adapter (CNA) replace in a server, and what key benefit does this provide?

- A. It replaces the storage controller; benefit is increased cache capacity
 - B. It replaces both HBAs and NICs with a single adapter; benefit is reduced adapter/cable/switch count and offloaded FCoE processing
 - C. It replaces the RAID controller; benefit is faster RAID rebuild
 - D. It replaces the Fibre Channel Forwarder; benefit is improved zoning
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7. (Medium) [Ch.7] In a Unified NAS architecture, what provides connectivity for both file-level (NAS) clients and block-level (iSCSI/FC) hosts?

- A. Separate, independently managed storage arrays for each access type
 - B. Shared storage controllers that connect to both NAS heads and provide iSCSI/FC ports
 - C. A dedicated FCoE switch only
 - D. The Control Station alone
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8. (Hard) [Ch.7] What is the recommended best practice when combining NAS and SAN workloads on the same storage system, regarding disk separation?

- A. Always combine NAS and SAN workloads on the same disks for efficiency
 - B. Separate NAS and SAN disks because NAS workloads (typically random, small I/O) can disrupt SAN sequential workloads
 - C. NAS and SAN should never share the same storage array under any circumstances
 - D. Disk separation is irrelevant; only cache size matters
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9. (Medium) [Ch.4] What term describes presenting hosts with more total LUN capacity than is physically available in the storage array, enabled by virtual provisioning?

- A. RAID expansion
 - B. LUN masking
 - C. Oversubscription
 - D. Cache vaulting
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10. (Hard) [Ch.4] In a high-end (active-active) storage array architecture, how can a host perform I/O to its LUN?

- A. Only through the single controller that owns the LUN
 - B. Through any of the available controllers
 - C. Only through a dedicated backup controller
 - D. Only via the front-end port directly attached to the disk
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11. (Medium) [Ch.5] Which FC class of service guarantees in-order frame delivery, uses end-to-end credit flow control, and provides a dedicated connection — but with poor bandwidth utilization?

- A. Class 1
 - B. Class 2
 - C. Class 3
 - D. Class F
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12. (Hard) [Ch.5] In FC zoning, why does the fabric controller send Registered State Change Notifications (RSCNs) only to nodes within an affected zone (rather than to all nodes) when zoning is configured?

- A. To increase the security of the WWN database
 - B. To limit unnecessary fabric-management traffic that would otherwise impact host-to-storage data traffic in a large fabric
 - C. To enable NPIV
 - D. To reduce the number of available FC addresses
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13. (Medium) [Ch.6] What is the function of the Fibre Channel Forwarder (FCF) within an FCoE switch?

- A. It manages VLAN tagging exclusively
 - B. It encapsulates FC frames (from the FC port) into FCoE frames, and de-encapsulates FCoE frames back into FC frames
 - C. It performs RAID parity calculations
 - D. It serves as the iSCSI target
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14. (Hard) [Ch.6] Why must FCoE use jumbo frames to maintain good performance?

- A. Because Ethernet's default frame payload (1,500 bytes) is smaller than a typical FC frame payload (2,112 bytes), risking the FC frame being split across two Ethernet frames
 - B. Because jumbo frames are required for VLAN tagging
 - C. Because jumbo frames eliminate the need for a CNA
 - D. Because jumbo frames are mandated by the BB_Credit mechanism
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15. (Medium) [Ch.7] What is the purpose of link aggregation in a NAS environment?

- A. To increase the physical distance supported by Ethernet cabling
- B. To combine two or more network interfaces into a single logical interface for higher throughput, load balancing, and transparent failover

- C. To convert NFS traffic into CIFS traffic
 - D. To enable jumbo frame support
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16. (Hard) [Ch.4] Which cache data protection mechanism addresses the risk of data loss specifically during an EXTENDED power failure, where battery power alone is insufficient?

- A. Cache mirroring
 - B. Cache vaulting (using dedicated vault drives)
 - C. Write-through caching
 - D. LUN masking
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17. (Medium) [Ch.5] What characteristic distinguishes midrange storage systems (active-passive arrays) from high-end storage systems (active-active arrays)?

- A. Midrange systems support more front-end ports than high-end systems
 - B. In midrange (active-passive) systems, a host can perform I/O to a LUN only through the controller that owns that LUN, while the other path remains passive
 - C. Midrange systems do not support RAID
 - D. Midrange systems have larger cache than high-end systems
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18. (Hard) [Ch.6] What problem does Priority-based Flow Control (PFC) solve compared to the traditional Ethernet PAUSE frame mechanism?

- A. PFC eliminates the need for any flow control
 - B. PFC allows pausing of a single traffic class/priority on a virtual lane without pausing the entire physical link, unlike the traditional PAUSE frame which affects the whole link
 - C. PFC increases the MTU size automatically
 - D. PFC replaces TCP windowing
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19. (Medium) [Ch.7] What does the term "scale-out NAS" specifically enable, that scale-up (unified/gateway) NAS does not?

- A. The ability to add CPUs and memory to an existing NAS head
 - B. The ability to add nodes to a clustered architecture for near-unlimited, non-disruptive scalability of performance and capacity together
 - C. The ability to use CIFS instead of NFS
 - D. The ability to bypass the need for an IP network
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20. (Hard) [Ch.5] Why is single HBA zoning considered an industry best practice, despite requiring more zones and administrative effort in a large fabric?

- A. It increases the number of available FC addresses
- B. It eliminates unnecessary host-to-host interaction and minimizes RSCNs, improving FC SAN performance and easing troubleshooting
- C. It is required by the FC-0 layer specification
- D. It automatically enables NPIV

Answer Key & Explanations — Section A

Q	Answer	Explanation
1	B	MRU frees the most recently accessed pages, the opposite of LRU, based on the assumption that recently used data may not be needed again for a while.
2	C	Forced flushing occurs when cache hits 100% capacity (typically during a large I/O burst); the system prioritizes flushing by allocating more resources, significantly impacting response time.
3	B	The core tier (composed of enterprise directors) ensures high fabric availability; all storage devices connect to the core, while edge-tier switches (connected to hosts) link to the core via ISLs.
4	B	Hop count is the total number of ISLs a packet traverses between source and destination; best practice keeps host-to-storage hops at one in core-edge designs to minimize transmission delay.
5	B	FCoE's key benefit is I/O consolidation — converging LAN and SAN traffic over one physical infrastructure using CNAs, reducing the number of adapters, cables, and switches needed.
6	B	A CNA replaces both the HBA and NIC, consolidating FC and Ethernet traffic into one adapter; it also offloads FCoE protocol processing from the server CPU.
7	B	In unified NAS, NAS heads connect to shared storage controllers that also provide iSCSI and FC ports, enabling both file and block access from the same consolidated platform.
8	B	NAS workloads are typically random with small I/O sizes; combining them with sequential SAN workloads can be disruptive, so separating NAS and SAN disks is recommended.
9	C	Oversubscription is presenting more total logical capacity to hosts than physically exists in the array, made possible by virtual provisioning's on-demand allocation.
10	B	In active-active (high-end) arrays, a host can perform I/O through ANY available controller, unlike active-passive (midrange) arrays where only the owning controller's path is active.
11	A	Class 1 provides a dedicated connection with end-to-end credit flow control and guaranteed in-order, acknowledged delivery — but no multiplexing means poor bandwidth utilization.
12	B	Without zoning, an RSCN goes to ALL nodes in the fabric for any change, generating significant fabric-management traffic in large fabrics; zoning limits RSCNs to only the affected zone's members.
13	B	The FCF's function is to encapsulate FC frames received from the FC port into FCoE frames, and de-encapsulate FCoE frames from the Ethernet Bridge back into FC frames.
14	A	A standard Ethernet frame's default payload is 1,500 bytes, while a typical FC frame payload is up to 2,112 bytes; without jumbo frames, an FC frame would be split across two Ethernet frames, harming performance.

Q	Answer	Explanation
15	B	Link aggregation combines multiple NICs into one logical interface, enabling higher throughput, load balancing/sharing, and transparent path failover if a connection is lost.
16	B	Cache vaulting dumps cache contents to dedicated vault drives during an extended outage, since batteries cannot sustain writing all data to disk over a long period; cache mirroring instead protects against a memory card failure, not an extended outage.
17	B	Midrange (active-passive) arrays route I/O for a given LUN only through its owning controller — the other path remains passive — unlike high-end (active-active) arrays where any controller can serve I/O.
18	B	PFC creates 8 independent virtual lanes on one physical link, so a specific priority/class (e.g., FCoE traffic) can be paused without pausing all traffic, unlike the standard Ethernet PAUSE frame which halts the entire link.
19	B	Scale-out NAS uniquely allows adding nodes to a cluster for combined performance AND capacity scaling without downtime, unlike scale-up approaches (unified/gateway) which are limited by a single device's housing capacity.
20	B	Single HBA zoning (one HBA port + its storage targets per zone) eliminates unneeded host-to-host visibility and reduces RSCN volume, which improves performance and simplifies troubleshooting, even though it increases the total zone count.

SECTION B: Structural Questions (30 Marks)

Question 1 (10 Marks)

(a) Describe cache mirroring and cache vaulting as cache data protection mechanisms, explaining what specific risk each addresses. **[6 marks]**

(b) What is "cache coherency" and why does it become a concern in a cache mirroring implementation? **[4 marks]**

Model Answer

(a) Cache Mirroring and Vaulting [6 marks — 3 marks each]:

Cache mirroring addresses the risk of a cache memory card/module failure. Every write to cache is stored in two different memory locations on two independent memory cards. If one card fails, the data remains safe in the mirrored location and can still be committed to disk. Only writes (not reads) are mirrored, since reads can always be re-staged from disk if needed — this conserves cache capacity.

Cache vaulting addresses the risk of an extended power failure. Short outages can be bridged using battery power (to keep memory powered, or to flush cache to disk), but batteries cannot sustain power long enough to commit large volumes of data across numerous disks during an extended outage. Instead, the entire cache

content is dumped onto a dedicated set of physical disks called vault drives; when power is restored, this data is written back to cache and then to its intended disk locations.

(b) Cache coherency [4 marks]:

Cache coherency means that data held in the two mirrored cache locations must remain identical at all times. This becomes a concern in cache mirroring because any write must be correctly propagated and kept synchronized across both memory cards — if the two copies were to diverge, a subsequent read or recovery operation could retrieve stale or incorrect data. Ensuring this coherency is the responsibility of the array's operating environment.

Question 2 (10 Marks)

(a) Compare full mesh and partial mesh FC SAN topologies in terms of scalability, hop count, and typical deployment size. **[6 marks]**

(b) Explain the concepts of "fan-out" and "fan-in" in an FC SAN, and identify what governs each ratio. **[4 marks]**

Model Answer

(a) Full Mesh vs Partial Mesh [6 marks]:

Aspect	Full Mesh	Partial Mesh
Connectivity	Every switch connected to every other switch	Not all switches directly interconnected
Hop count	Maximum of one ISL/hop for host-to-storage traffic	Several hops/ISLs may be required
Scalability	Limited — as switches increase, more ports are consumed by ISLs, reducing ports available for node connectivity	More scalable than full mesh
Typical deployment	Small number of switches (up to ~4) servicing highly localized traffic	Larger deployments, but requires careful host/storage placement to avoid ISL overload

(b) Fan-out and Fan-in [4 marks]:

Fan-out refers to the number of server (host) ports that connect to communicate with a single storage port. For example, four servers connecting to one storage port yields a fan-out ratio of 4. The fan-out ratio is governed by the **front-end processing capability of the storage system**, and the product vendor typically specifies the maximum supported ratio.

Fan-in refers to the number of storage ports that a single server port uses. Similar to fan-out, the fan-in ratio is constrained by the **capability of the host-bus adapter (HBA)**.

Question 3 (10 Marks)

(a) Explain the I/O consolidation benefit of FCoE by describing the data center infrastructure before and after FCoE deployment. **[6 marks]**

(b) What is the role of the MTU (Maximum Transmission Unit) setting in NAS performance, and why are jumbo frames beneficial? **[4 marks]**

Model Answer

(a) FCoE I/O Consolidation [6 marks]:

Before FCoE: Data centers maintain two discrete physical network infrastructures — an Ethernet/IP network (accessed via NICs) for TCP/IP traffic, and an FC network (accessed via HBAs) for block-level storage traffic. Each server is typically equipped with multiple NICs (2–4) and redundant HBAs, requiring large numbers of adapters, cables, and switches across hundreds of servers — increasing complexity, and the cost of power, cooling, and floor space.

After FCoE: A Converged Network Adapter (CNA) in each server replaces both HBAs and NICs, consolidating both IP and FC traffic over a single physical interface connected to an FCoE switch. This dramatically reduces the number of required adapters, cables, and switches, lowering cost and management overhead while preserving the ability to communicate with both Ethernet/IP and native FC SAN infrastructure (the FCoE switch contains a Fibre Channel Forwarder to bridge to traditional FC SANs).

(b) MTU and Jumbo Frames [4 marks]:

The MTU setting determines the size of the largest packet that can be transmitted without fragmentation. The default Ethernet MTU is 1,500 bytes. **Jumbo frames** allow transmission of Ethernet frames with an MTU greater than 1,500 bytes (commonly 9,000 bytes). Because servers transmit and receive larger frames more efficiently under heavy traffic, jumbo frames reduce the number of frames needed to transfer the same amount of data, reducing per-frame protocol overhead, conserving raw network bandwidth, and smoothing sudden I/O bursts — all of which improve NAS performance.

SECTION C: Essay Questions (50 Marks)

Essay Question 1 (25 Marks)

Discuss the architectural design choices that determine performance, availability, and scalability across intelligent storage systems and Fibre Channel SAN fabrics. Your answer should address: high-end vs midrange storage array architectures, FC SAN topology choices (mesh and core-edge), and the role of zoning and classes of service in managing fabric behavior.

Model Answer

Introduction [2 marks]

The performance, availability, and scalability of an end-to-end storage infrastructure depend on architectural decisions made at multiple layers: how the storage array itself routes I/O internally, how switches are

interconnected to form a fabric, and how that fabric is logically segmented and governed. This essay examines these design choices and their trade-offs.

High-End vs Midrange Storage Array Architectures [6 marks]

Storage systems generally fall into two categories. **High-end storage systems** (active-active arrays) are designed for large enterprise applications with numerous controllers and large cache memory. In an active-active configuration, a host can perform I/O to a given LUN through **any** of the available controllers, maximizing path flexibility, load balancing, and fault tolerance — critical for mission-critical, unpredictable workloads.

Midrange storage systems (active-passive arrays) are best suited for small-to-medium enterprise applications, offering lower cost. In an active-passive configuration, a host can perform I/O to a LUN **only** through the controller that owns that LUN — the alternate path remains passive with no I/O activity, serving purely as a failover route. Midrange arrays typically have fewer front-end ports, less cache, and lower capacity than high-end systems, but they still provide high redundancy, predictable performance, and array-based replication support — making them well suited to workloads with predictable demand profiles. The distinction between these architectures has become increasingly blurred over time as midrange systems have adopted more high-end capabilities.

FC SAN Topology Choices: Mesh vs Core-Edge [9 marks]

Mesh topology comes in two forms. In a **full mesh**, every switch connects directly to every other switch, guaranteeing a maximum of one ISL hop for any host-to-storage communication — ideal for small deployments (up to ~4 switches/directors) with localized traffic. However, as switch count grows, an increasing proportion of switch ports must be dedicated to ISLs rather than node connectivity, limiting scalability. A **partial mesh** sacrifices the guaranteed single-hop property — several hops may be required — in exchange for improved scalability, though without careful host/storage placement, ISLs risk becoming overloaded with aggregated traffic.

Core-edge fabric topology instead organizes switches into two tiers. The **edge tier** (often less expensive switches) connects to hosts and provides an economical way to add connectivity; edge switches are not directly connected to each other. The **core tier** (typically high-end directors) ensures high availability, and all storage devices connect to the core, so that essentially all host-to-storage traffic traverses exactly one ISL hop (the best practice is to keep this hop count at one). Core-edge fabrics can be scaled by adding more edge switches (single-core or dual-core variants) or by adding more core switches — though beyond a certain scale, maintaining ISLs from every core to every edge switch becomes impractical, necessitating a transition to a **compound/complex core-edge design**.

The fundamental trade-off is: mesh topologies offer simplicity and minimal latency at small scale but degrade in port efficiency as they grow; core-edge topologies offer a more deterministic, easily analyzed traffic pattern (since each switch port serves either storage or hosts exclusively) and scale more gracefully, at the cost of architectural complexity when extended to compound designs.

Zoning and Classes of Service [6 marks]

Zoning is the mechanism by which node ports within a fabric are logically segmented into groups that can communicate only with members of the same group. Beyond access control, zoning serves a critical performance function: it limits the propagation of Registered State Change Notifications (RSCNs). Without zoning, any fabric change triggers an RSCN to every node in the fabric, generating substantial management

traffic in large fabrics that can impact actual host-to-storage data traffic. With zoning, RSCNs are confined to the affected zone. **WWN zoning** offers superior operational flexibility versus **port zoning** because it is tied to a device's static WWN rather than its physical switch port, allowing devices to be moved without reconfiguring zones. **Single HBA zoning** (one HBA paired with its specific targets per zone) is the recommended best practice, minimizing unnecessary host-to-host visibility and RSCN volume, despite the administrative overhead of managing more, smaller zones.

Classes of Service further shape fabric behavior at the transport level: Class 1 provides a dedicated, guaranteed in-order connection (poor bandwidth utilization due to no multiplexing), Class 2 offers acknowledged, non-dedicated, multiplexed delivery without order guarantees, and Class 3 (most common) drops acknowledgment entirely for higher bandwidth utilization, relying on upper layers for reliability.

Conclusion [2 marks]

These architectural layers — array controller design, fabric topology, zoning, and class of service — are not independent choices; they compound. An active-active array connected through a well-designed core-edge fabric with WWN-based single-HBA zoning delivers a level of performance, availability, and manageability that no single design decision could achieve alone, illustrating why enterprise storage architecture demands holistic, end-to-end design thinking.

Essay Question 2 (25 Marks)

Discuss the technologies that enable convergence and extension of storage networking beyond the traditional, localized Fibre Channel fabric. Your answer should address: FCIP for SAN extension, FCoE for LAN/SAN convergence (including its enabling lossless Ethernet technologies), and how unified and scale-out NAS architectures address the scalability and convergence needs of file-based storage.

Model Answer

Introduction [2 marks]

Traditional Fibre Channel SAN excels at high-performance, localized block storage but is constrained by distance limitations and requires dedicated, costly infrastructure. Three complementary technologies — FCIP, FCoE, and modern NAS architectures — address different dimensions of this constraint: geographic extension, infrastructure convergence, and file-storage scalability, respectively.

FCIP: Extending SAN Across Distance [6 marks]

FCIP (Fibre Channel over IP) is a tunneling protocol that interconnects geographically distributed FC SAN islands over existing IP infrastructure, avoiding the cost of dedicated long-distance FC links. An FCIP gateway is connected to each fabric via standard FC connection; one gateway encapsulates FC frames into IP packets for transmission across the IP network, while the gateway at the receiving end strips the IP wrapper and forwards native FC frames into the local fabric — with each fabric treating the gateways as Layer 2 fabric switches. This makes FCIP especially valuable for **disaster recovery (DR)** implementations, where data must be replicated to a remote site.

However, FCIP introduces dependencies on the underlying IP network's quality: insufficient bandwidth over extended distances can become a bottleneck, and because FCIP effectively unifies two fabrics, instability in the IP network (e.g., due to a WAN outage) can propagate into the SAN environment as a segmented fabric,

excessive RSCNs, or host timeouts. Vendors mitigate this by segregating FCIP traffic into separate virtual fabrics. Security is also a key concern since data traverses public IP infrastructure — IPsec is a commonly implemented safeguard.

FCoE: LAN/SAN Convergence [9 marks]

While FCIP extends FC over distance, FCoE addresses a different problem: the cost and complexity of maintaining **two parallel physical network infrastructures** (Ethernet for LAN traffic, FC for SAN traffic) within a single data center, each requiring its own adapters, cabling, and switches.

FCoE achieves convergence by mapping native FC frames directly onto Ethernet, replacing only the lowest two layers of the FC stack (FC-0 physical, FC-1 encoding) with Ethernet's physical and MAC layers, while preserving FC-2 through FC-4 unchanged. This is enabled by the **Converged Network Adapter (CNA)**, which integrates 10GbE, FC, and FCoE ASICs to replace both the HBA and NIC in a server, and by the **FCoE switch**, which contains both an Ethernet Bridge and a **Fibre Channel Forwarder (FCF)** that encapsulates/de-encapsulates FC frames to/from FCoE frames, inspecting each frame's Ethertype to determine whether to route it as native Ethernet traffic or hand it to the FCF for FC SAN delivery.

Because conventional Ethernet is inherently *lossy* (frames may be dropped under congestion) while FC guarantees lossless delivery via credit-based flow control, FCoE requires **Converged Enhanced Ethernet (CEE)**, also called lossless Ethernet, defined by the Data Center Bridging (DCB) task group. CEE's core technologies are: **Priority-based Flow Control (PFC)**, which divides one physical link into 8 independently pausable virtual lanes so FCoE traffic can be made lossless without pausing unrelated traffic classes; **Enhanced Transmission Selection (ETS)**, which allocates and dynamically reassigns bandwidth among traffic classes (LAN, SAN, IPC); **Congestion Notification (CN)**, which signals upstream senders to throttle transmission when downstream congestion is detected; and the **Data Center Bridging Exchange Protocol (DCBX)**, which negotiates and distributes these capabilities consistently across switches and adapters. FCoE additionally requires jumbo frame support, since a standard 1,500-byte Ethernet payload is smaller than the up-to-2,112-byte FC frame payload it must carry.

Unified and Scale-Out NAS: Converging and Scaling File Storage [6 marks]

Parallel to block-storage convergence, NAS architectures address convergence and scale for file-based storage. **Unified NAS** converges file (NFS/CIFS) and block (iSCSI/FC) access onto a single storage platform with shared storage controllers and a single management interface, eliminating the need for entirely separate file-serving and block-storage infrastructures. **Scale-out NAS** addresses a different axis — capacity and performance scalability — by clustering multiple nodes into a single logical file system that can grow by simply adding nodes, with data striped and rebalanced automatically across the cluster, all without downtime. This directly answers the demands of "Big Data" workloads that exceed what any single scale-up NAS device can handle.

Synthesis: A Common Theme [4 marks]

FCIP, FCoE, and unified/scale-out NAS each tackle a different axis of the same underlying challenge — making storage infrastructure more flexible, more cost-effective, and more scalable without sacrificing the reliability guarantees that justified investing in dedicated storage networking in the first place. FCIP extends FC's reach geographically over commodity IP links; FCoE collapses two separate physical infrastructures into one by re-engineering Ethernet to behave losslessly; and unified/scale-out NAS converges file and block access (or scales file access indefinitely) within the storage platform itself. Together they represent the broader industry trend toward infrastructure consolidation and convergence in the modern data center.

Conclusion [2 marks]

These technologies illustrate that "convergence" in storage networking takes multiple forms — across distance (FCIP), across protocol/physical layer (FCoE), and across access type (unified NAS) or scale (scale-out NAS) — each reducing cost and complexity while preserving the performance and reliability that distinguish dedicated storage networks from general-purpose IP infrastructure.

End of Practice Examination Set 2 of 4